

Advanced Animal Movement Modelling Training Workshop

Foundations of Movement Ecology

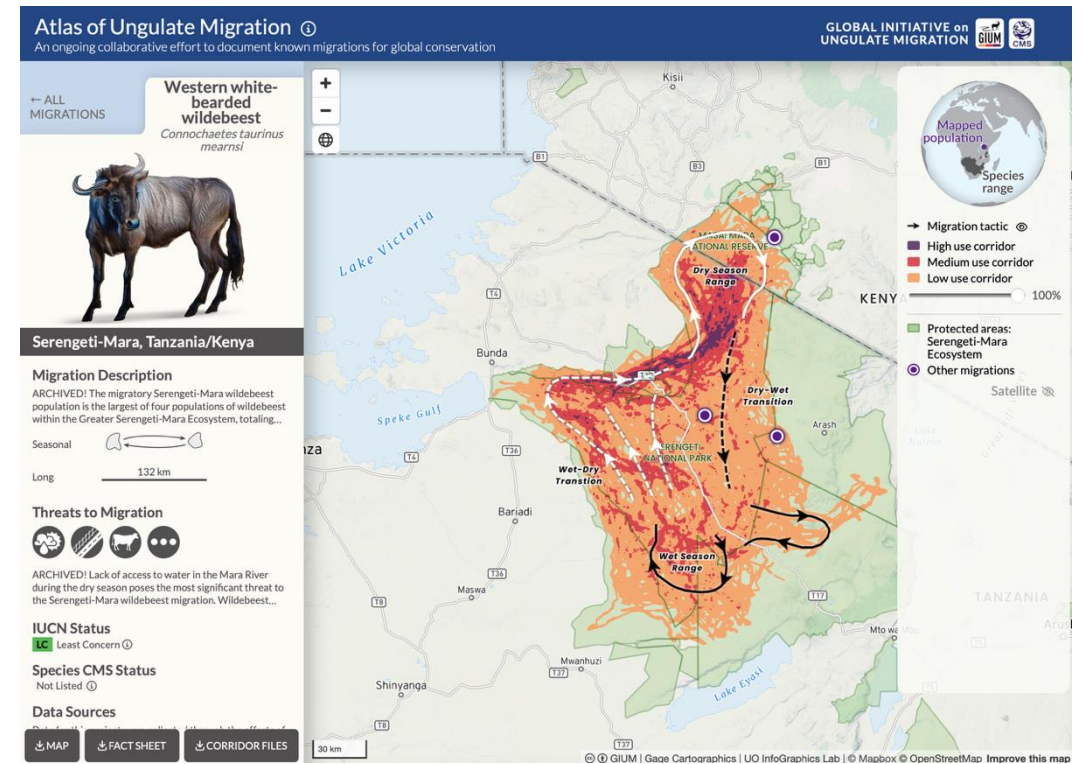
16 September 2026

Josh Cullen

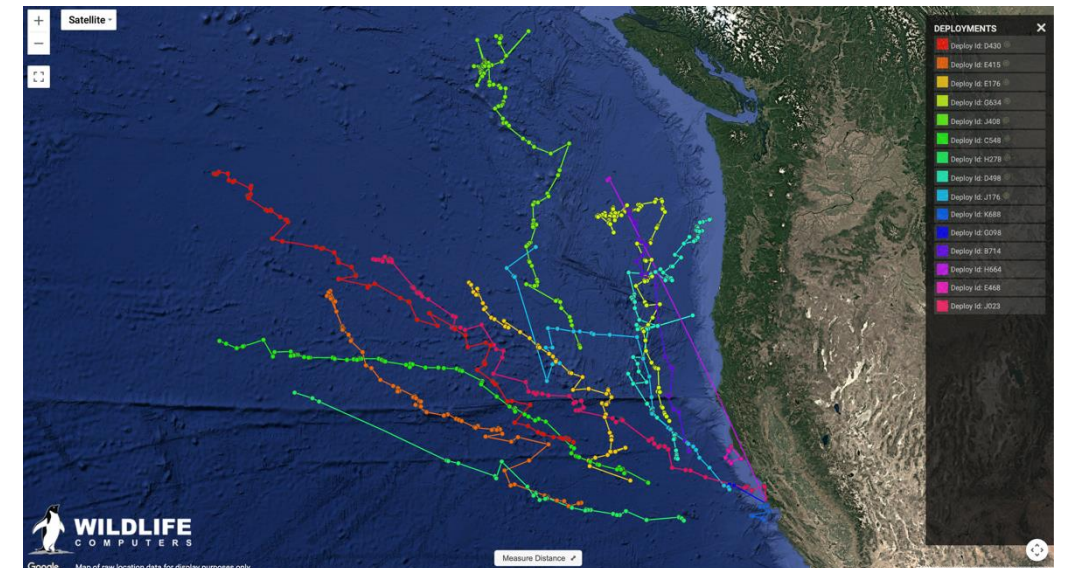


Background

- Movement is fundamental to the life history and fitness of organisms
- The study of animal movement via telemetry devices (e.g. GPS, Argos, VHF) has enabled a variety of detailed analyses



Beltran Lab: Post-breeding Northern Elephant Seals

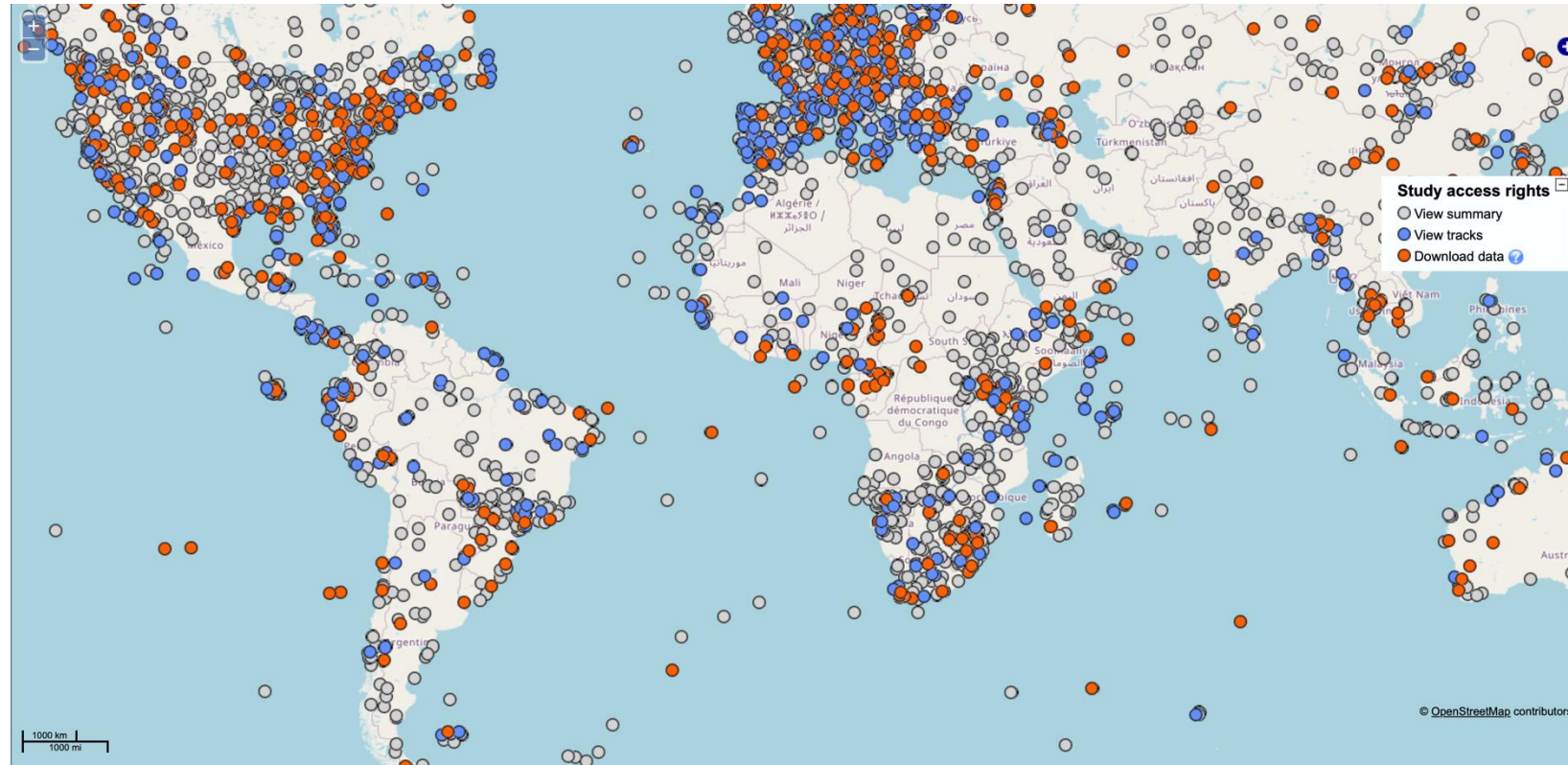


<https://www.cms.int/gium>

<https://my.wildlifecomputers.com/data/map/?id=699b11aa7a414b03f0a842b>

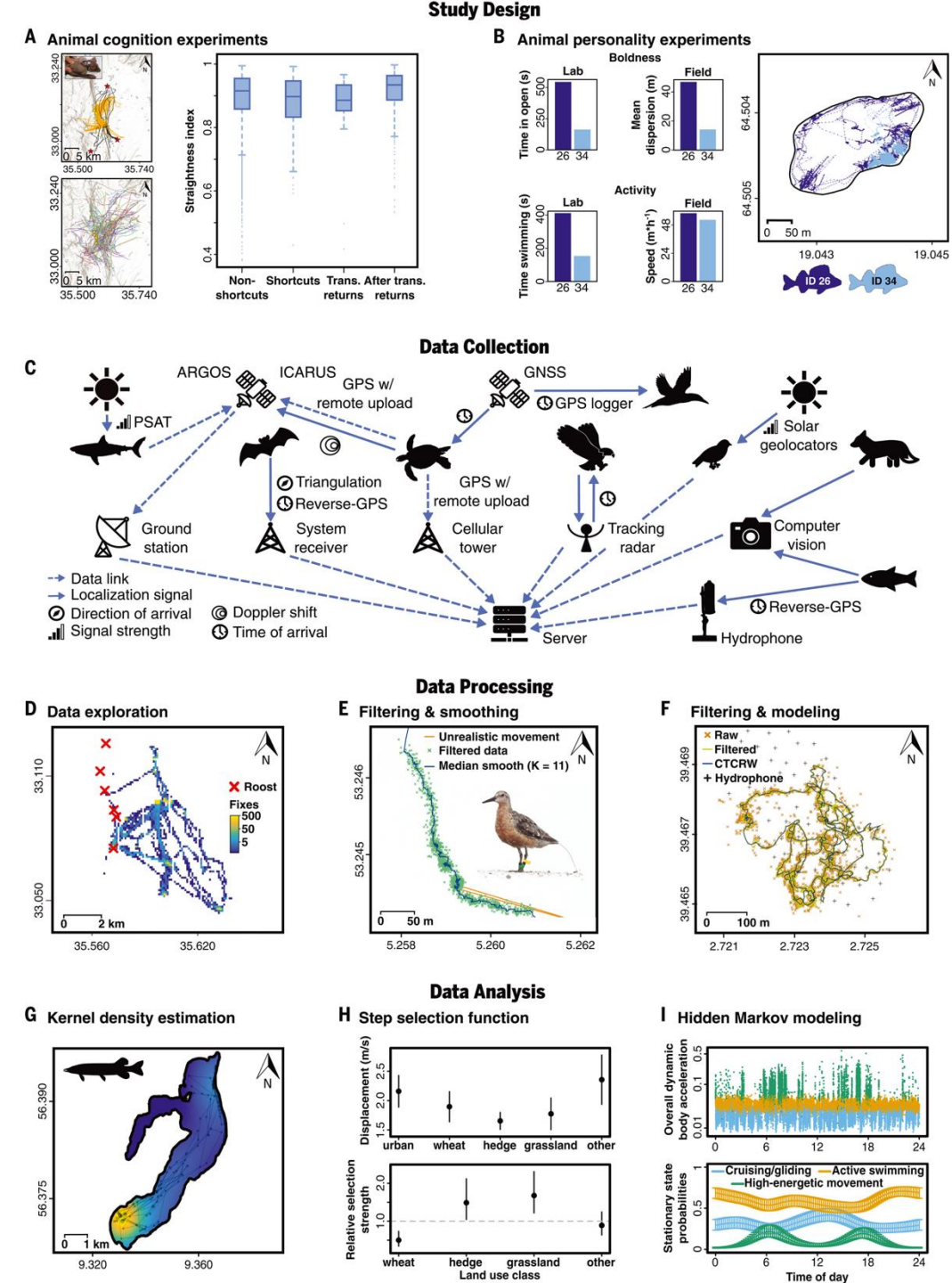
Background

Scientists and conservationists have collected large amounts of data on a wide array of species

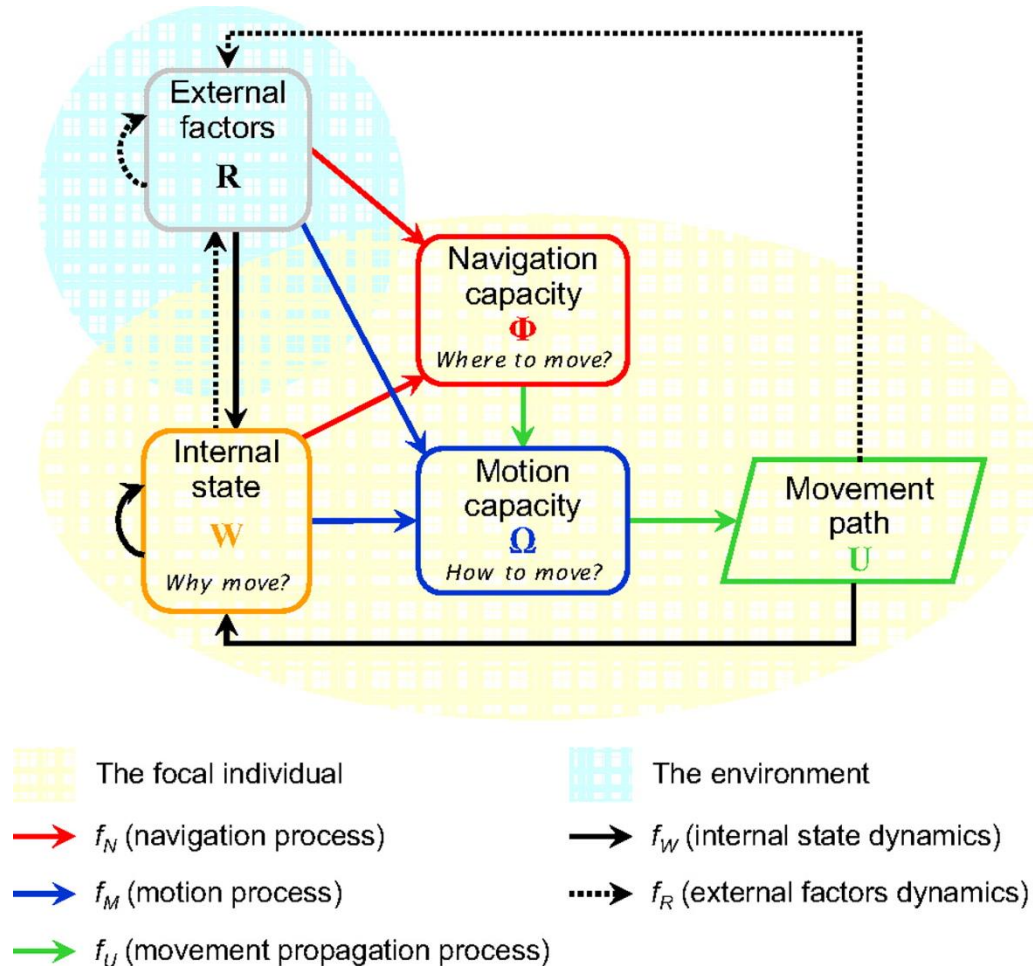


Background

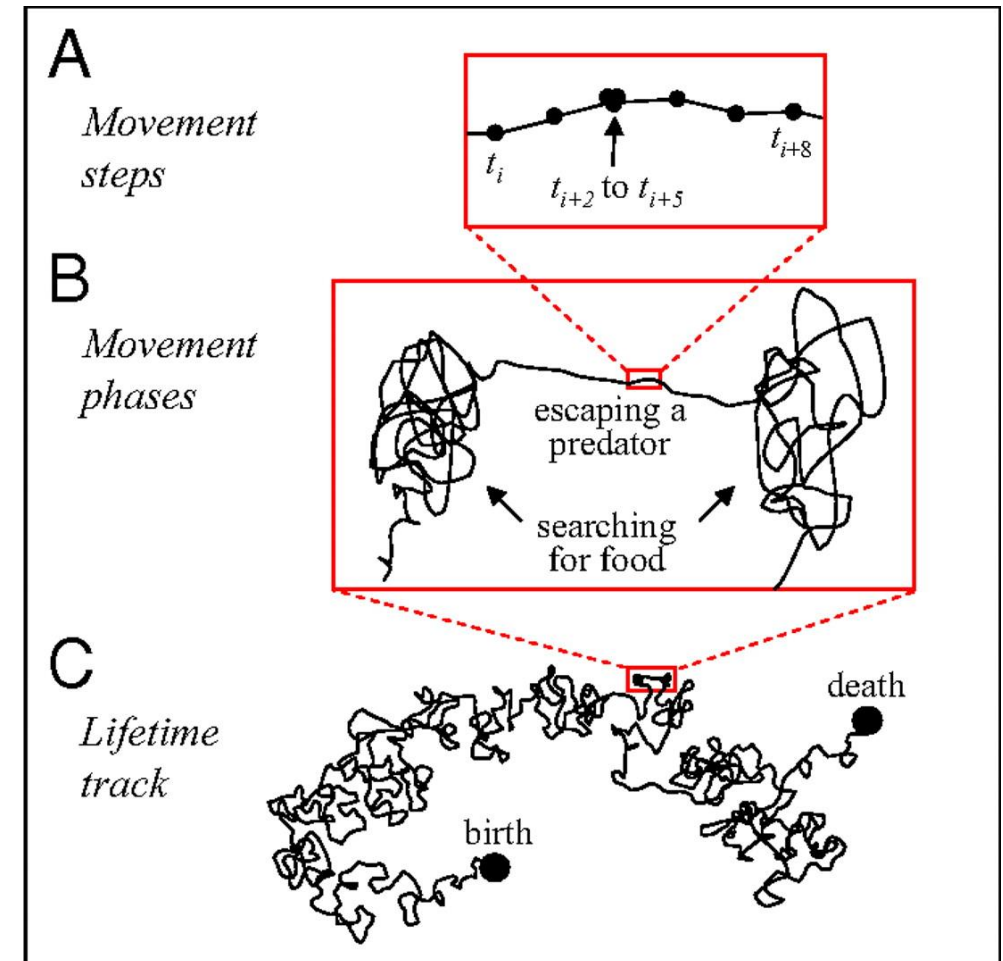
Often, researchers want to estimate space use, behavioral states, and species-habitat associations (SHAs) from animal tracks



Movement ecology paradigm



Increasing spatiotemporal scale



Big picture applications

Space use

- By an individual
- By a population
- Temporal patterns
- Relationship w/ protected areas
- Comparisons among life stages
- Effects of environ. and variability

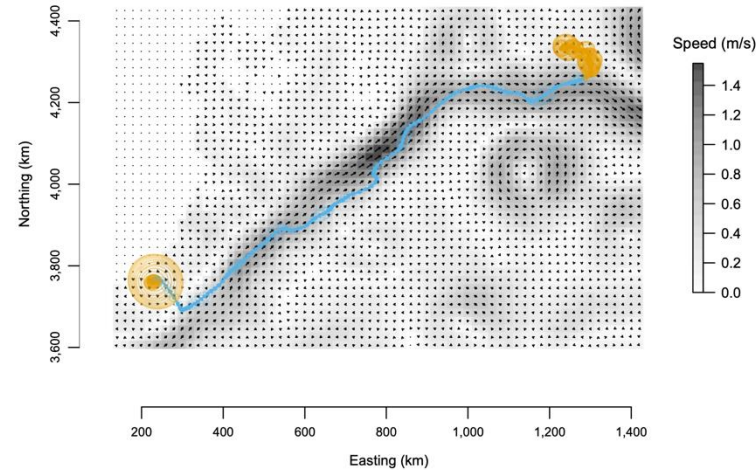
Behavioral states

- What is the animal doing?
- Activity budgets
- Identification of corridors or foraging grounds
- What factors drive behavioral shifts?

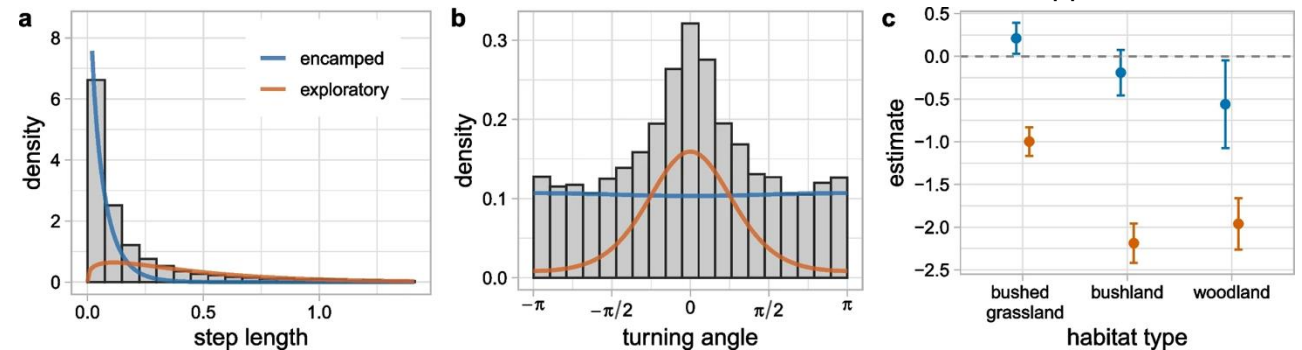
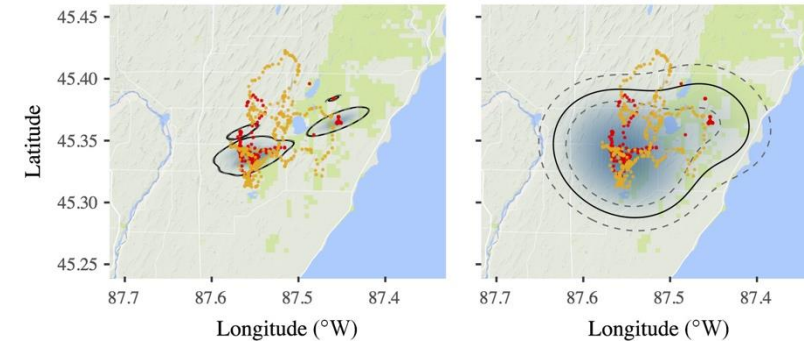
Species-habitat associations

- What habitat does the animal prefer?
- Does this differ by season/age/sex?
- Does this differ by behavioral state?
- How is this expected to be impacted by climate change?
- Can we use habitat preferences of some species of a community to inform habitat preferences of other species?

McClintock and Michelot 2018



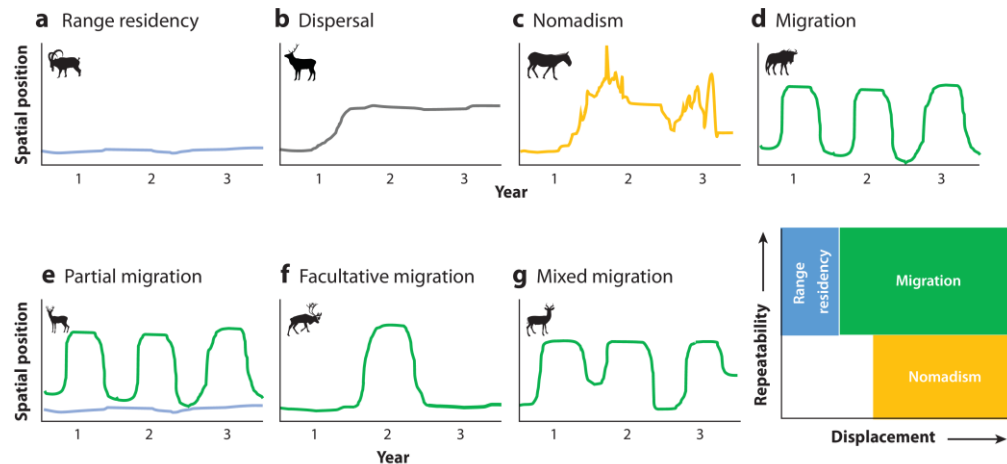
Noonan et al. 2019



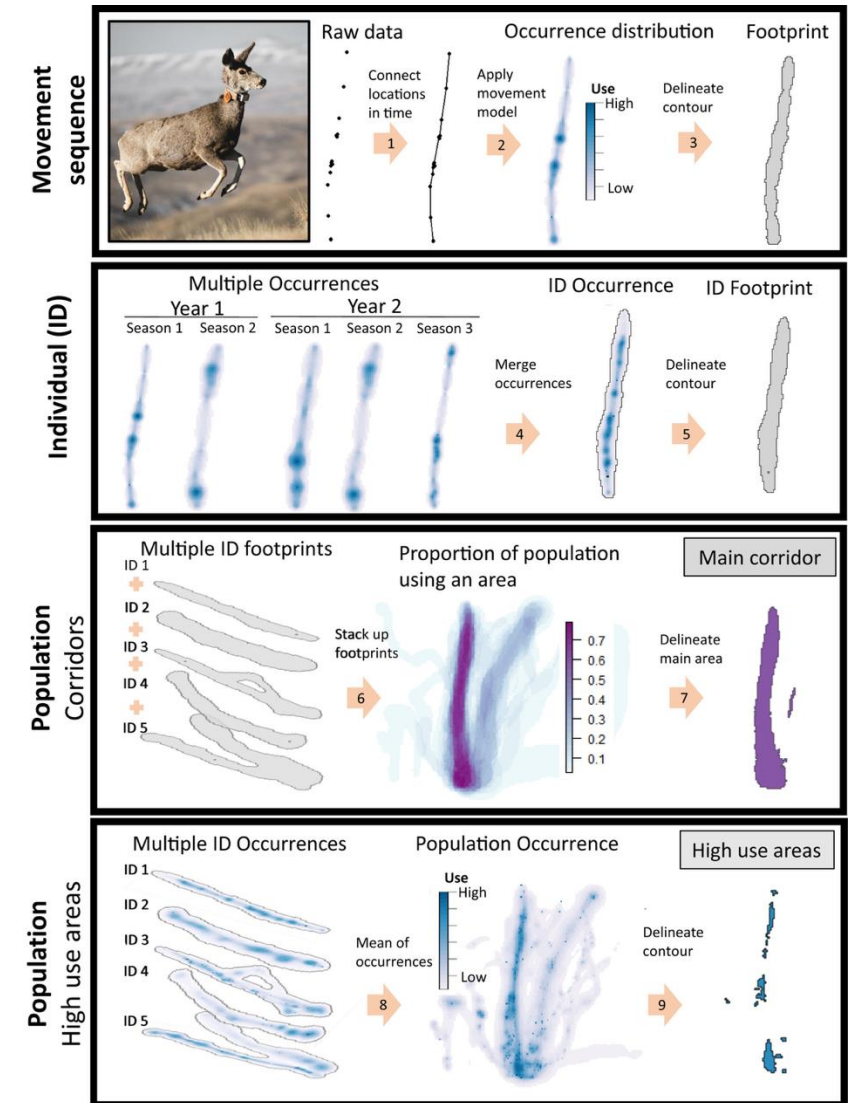
Space Use

Why estimate space use?

- Characterizing the spatiotemporal patterns of space use is fundamental to ecology
 - Where is the animal going?



Merkle et al. 2022



Concepts of space use

What does your space-use estimate mean?

Concept: Occurrence distributions

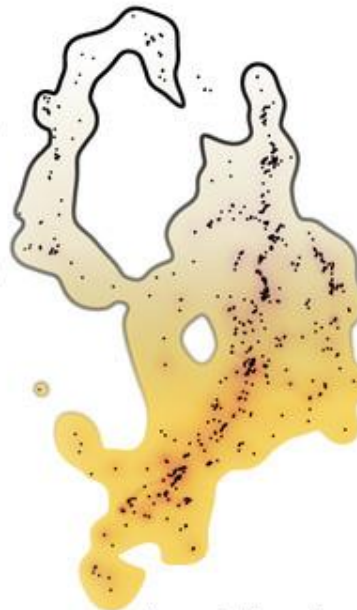
Estimating the space potentially used by animals during the sampling period, essentially estimates of uncertainty between known locations.

Distribution: Occurrence distribution

Contour: Confidence area

Predictive scope: Within sample interpolation

Example question: What potential movement corridors does species A use?



*e.g., dynamic Brownian
Bridge Movement
Models*

Range distributions

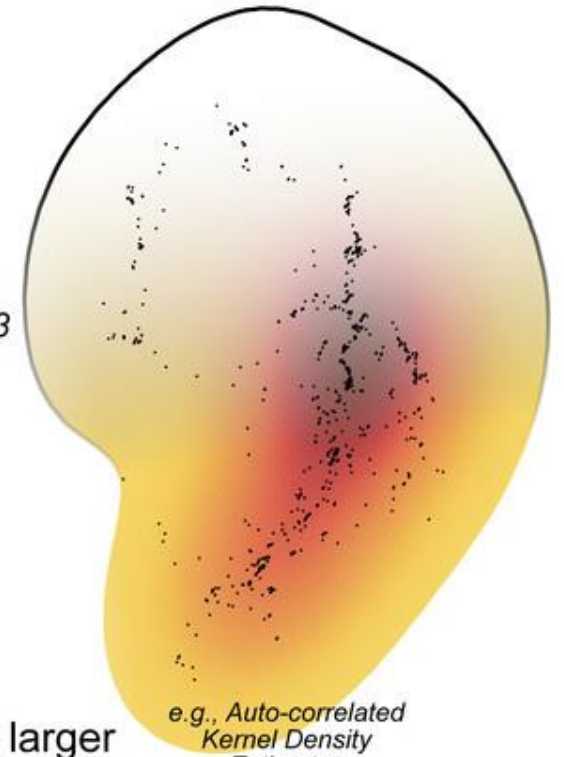
Estimating the space requirements of animals, corresponding to Burt's 1943 concept of home range.

Utilisation distribution

Home range

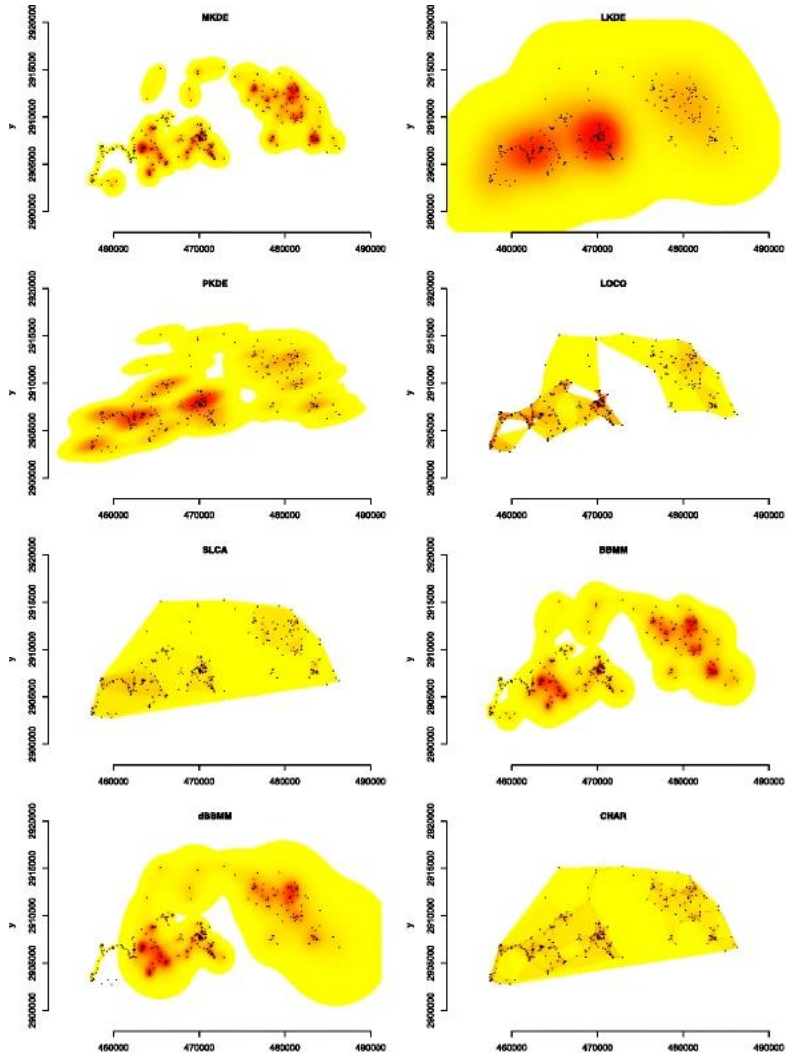
Beyond sample extrapolation

Does species A have a larger home range than species B?



*e.g., Auto-correlated
Kernel Density
Estimates*

Space use estimation



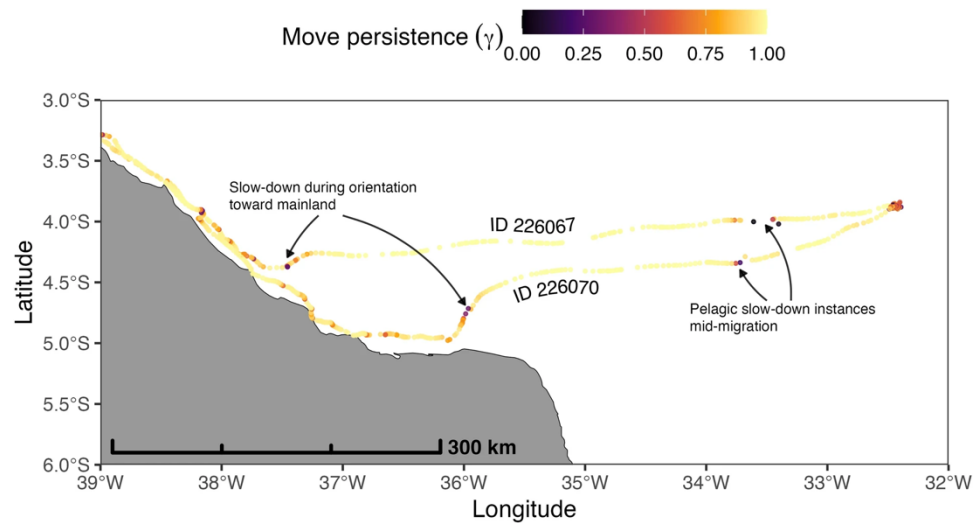
Walter et al. 2015

- Large number of methods for estimating space use
 - All of which result in quite different estimates
- Methods that continue to be frequently used include MCP, KDE, dBBMM, and AKDE

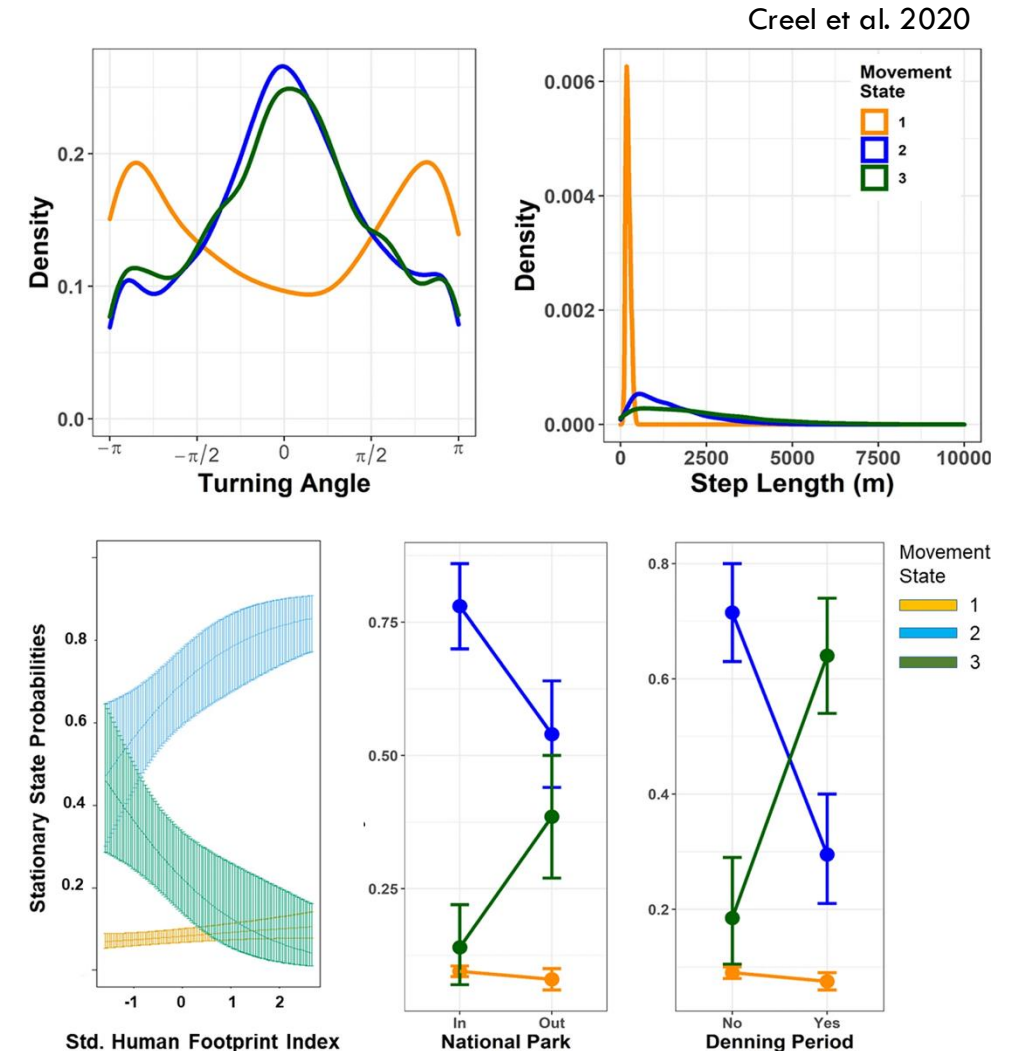
Behavioral States

Why estimate behavior?

- “Nothing in animal movement ecology makes sense except in the light of behavior” — Dobzhansky, 1973 (sort of)
 - What is the animal doing?

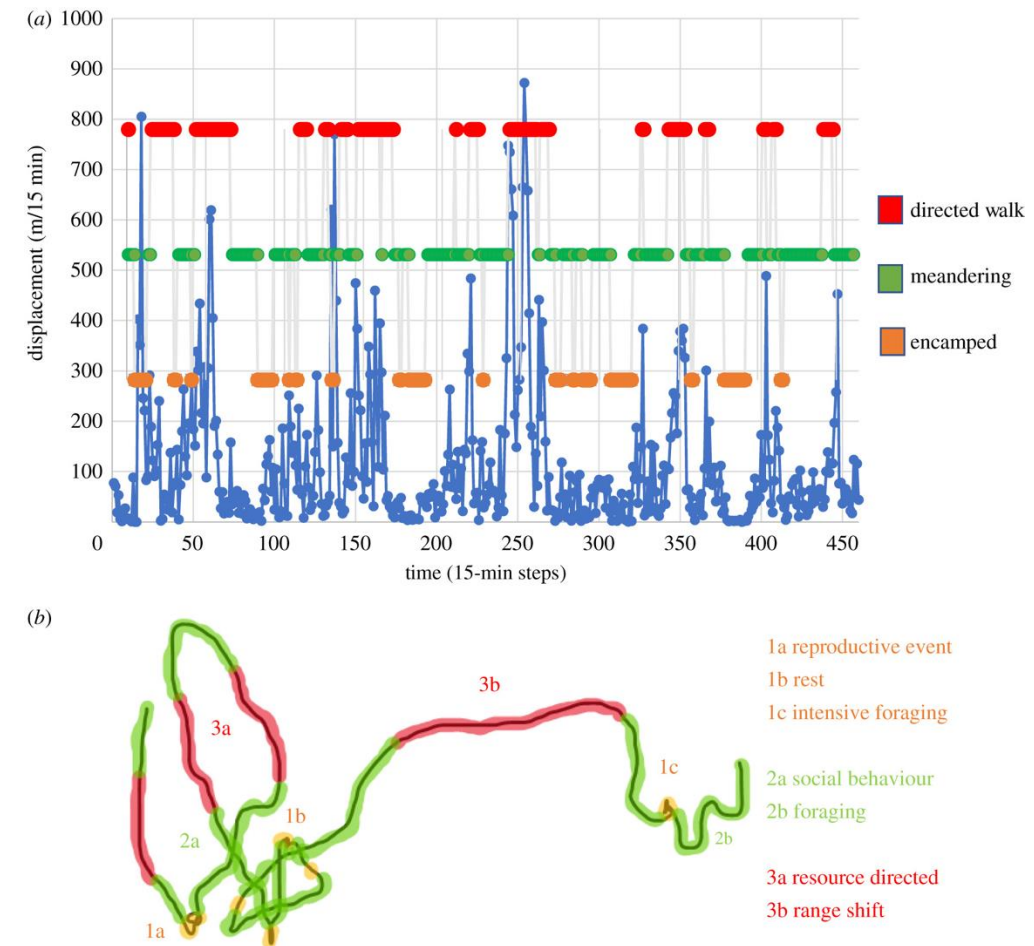


Cullen et al., 2025



What are latent behavioral states?

- In the scientific literature, you'll often see the term “latent behavioral states”
 - Latent —> in this case meaning “hidden”; since the true states are not directly observed, we can only infer them from the tracks
 - Behavioral states —> since we're not directly observing what the animals are doing, we can't define these as true “behaviors”



Behavioral state classification

40,000 ft view

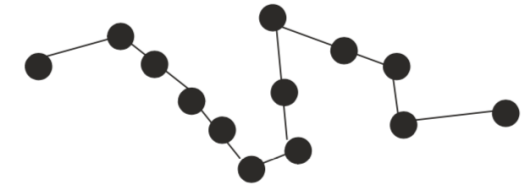
Table 1. Summary table of four broad categories of behavioural movement analysis methods. The four methods implemented in this paper and the most directly relevant references are bold faced. All of the entries in the last category can be considered multistate random walks, hidden Markov models or state space models

Category	Method	References
Metric-based	Fractal analysis	Fritz, Said & Weimerskirch (2003), Laidre <i>et al.</i> (2004) Nams & Bourgeois (2004); Tremblay, Roberts & Costa (2007)
	Tortuosity measures	Bovet & Benhamou (1988); Benhamou (2004)
Classification and segmentation	First passage time (FPT)	Fauchald & Tveraa (2003)
	Residence time (RT)	Barraquand & Benhamou (2008)
	Penalized contrasts	Lavielle (2005), Calenge (2006)
	Bayesian partitioning (BPM)	Calenge (2006)
Phenomenological time-series analysis	k-clustering	van Moorter <i>et al.</i> (2010)
	RT (segmentation step)	Barraquand & Benhamou (2008)
	Autocorrelation functions	Boyce <i>et al.</i> (2010)
Mechanistic movement modelling	Change point analysis (BCPA)	Gurarie, Andrews & Laidre (2009), Gurarie (2013)
	Wavelet	Kranstauber <i>et al.</i> (2012)
	Multistate random walk (MRW)	Morales <i>et al.</i> (2004)
	Ignoring location error	Forester <i>et al.</i> (2007), Langrock <i>et al.</i> (2012) Patterson <i>et al.</i> (2008), McClintock <i>et al.</i> (2012)
	Accounting for error	Jonsen <i>et al.</i> (2013), Breed <i>et al.</i> (2012)

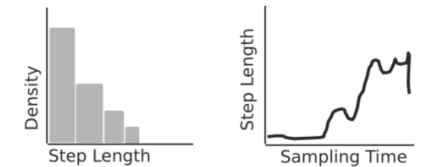
Actual Movement Path:



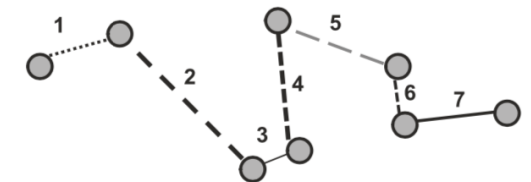
Step 1:



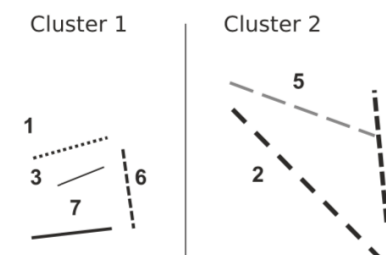
Step 2:



Step 3:



Step 4:

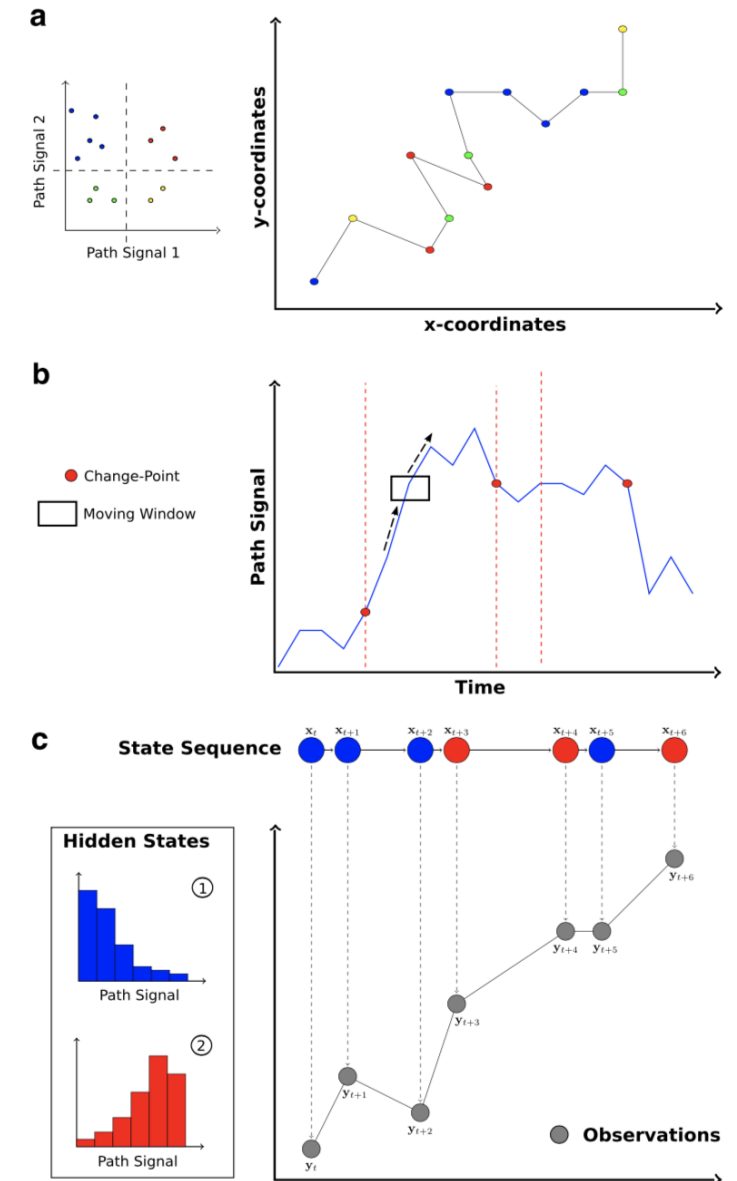


Behavioral state classification

40,000 ft view

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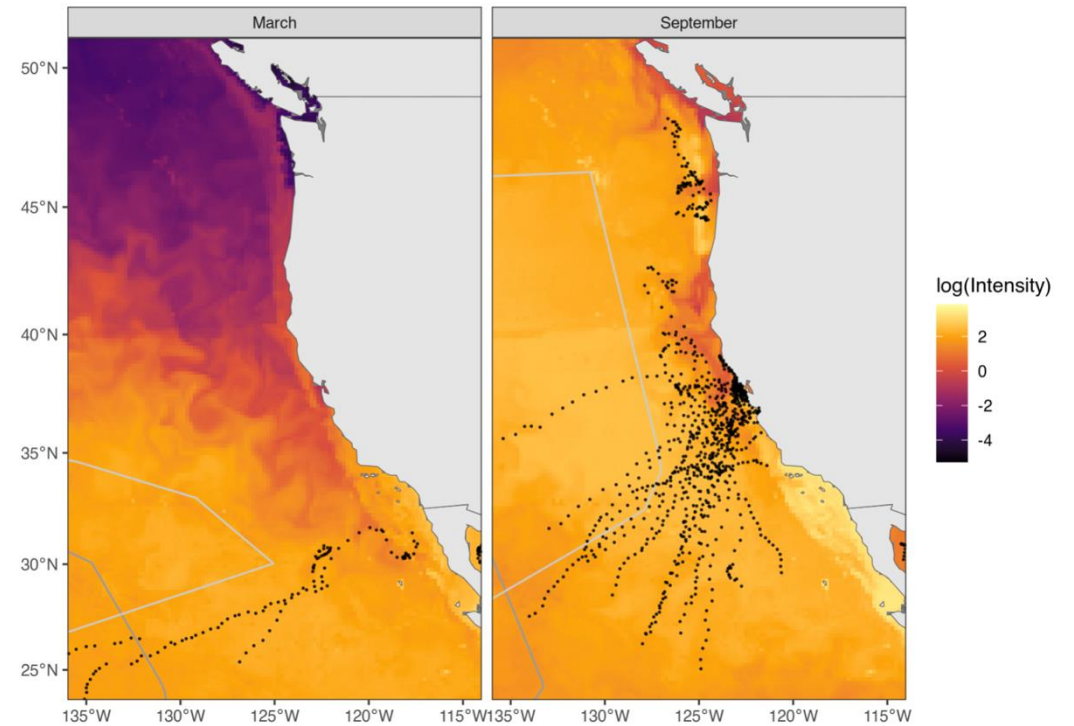
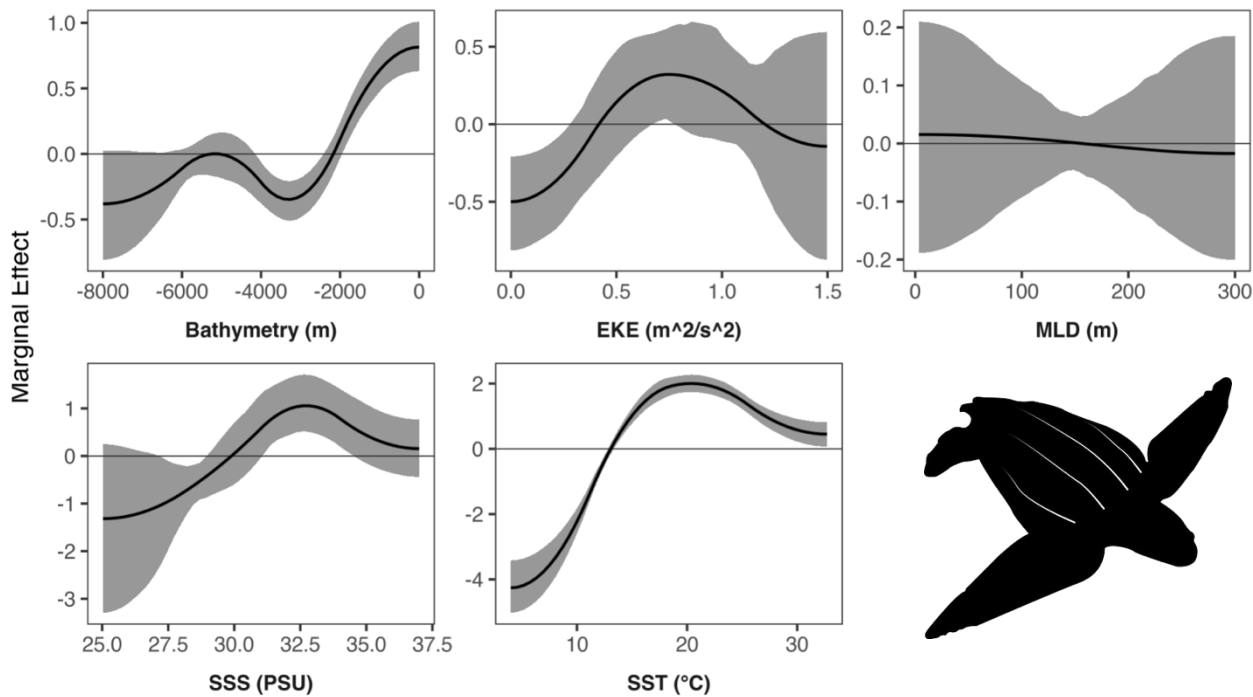
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Species-Habitat Associations

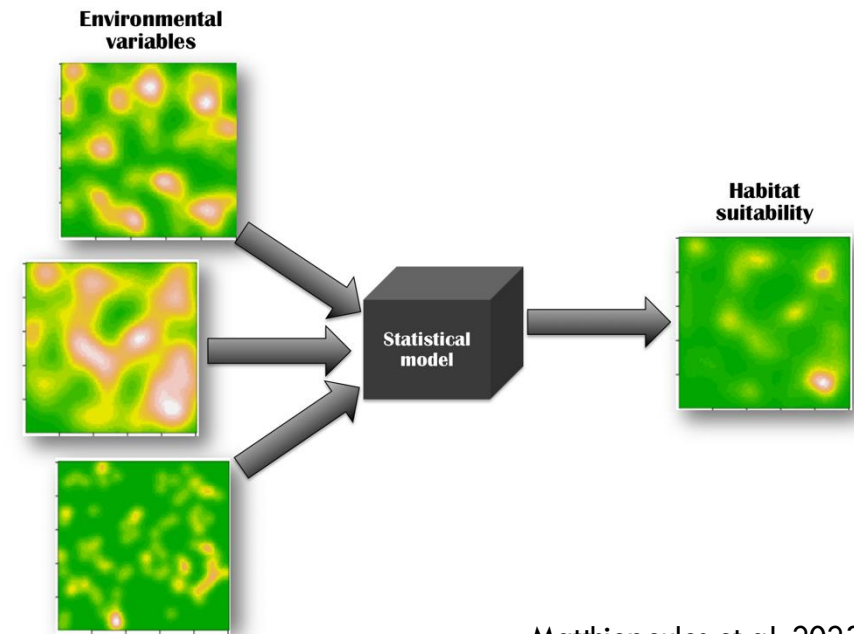
Why estimate SHAs?

- Allows us to estimate responses to the environment and predict species distributions
 - Why is the animal there? How many?



What are SHAs?

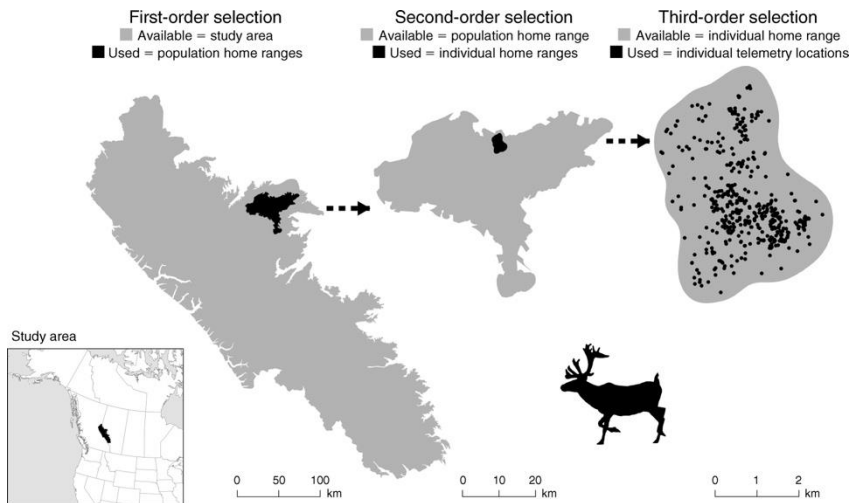
- Per Matthiopoulos et al. (2023):
 - *"A model that connects aspects of habitat (resources, conditions, risks) to particular observations about a species (recorded at the individual, group or population levels). Most often, the observations relate to species densities ... but can also relate to population fitness or changes in species density."*
- Often estimated via resource (RSF) or step selection functions (SSFs)
 - Each quantifies a different scale of selection
 - RSFs also often referred to as "habitat selection functions" to account for risks AND resources



Matthiopoulos et al. 2023

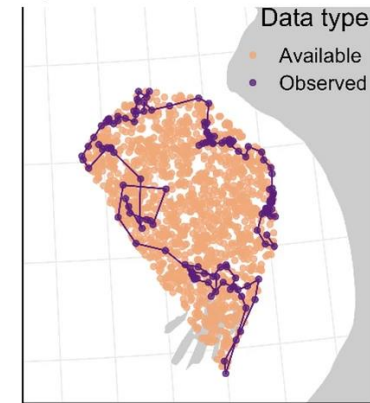
Evaluating different scales of selection

- SHAs can be estimated at multiple spatial scales of interest (per Johnson (1980))
- Different scales of SHAs can be estimated with RSFs, or RSFs and SSFs

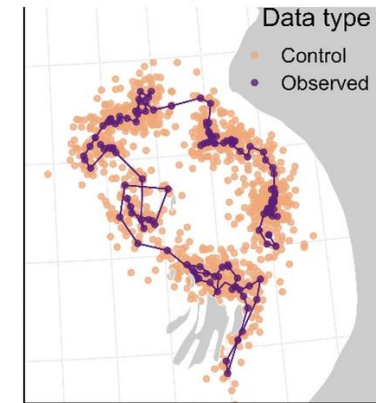


DeCesare et al. 2012

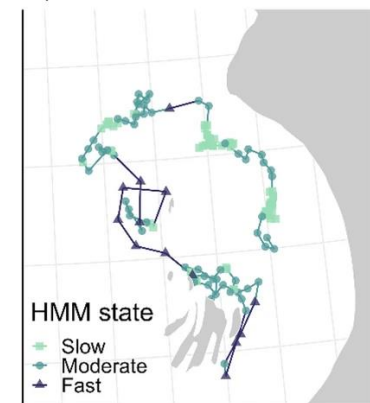
A) RSF samples



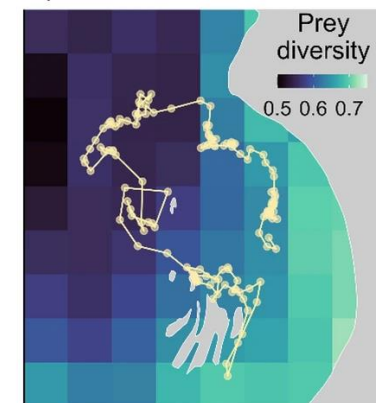
B) SSF samples



C) HMM decoded states



D) Seal track



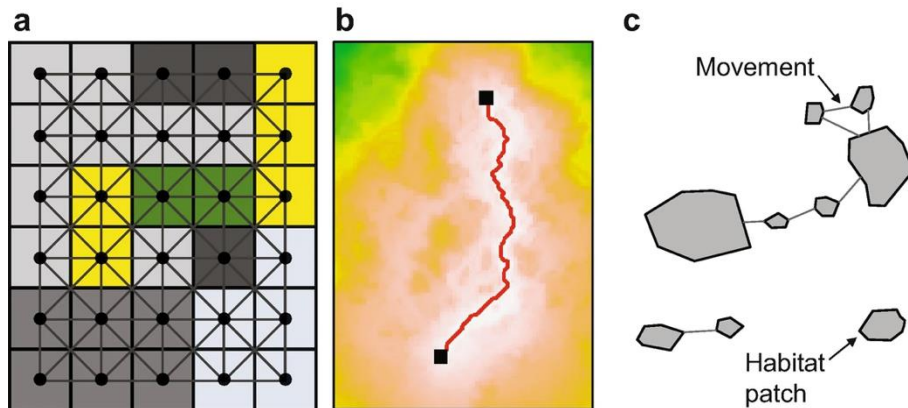
Florko et al. 2025

Bonus Topics

Connectivity, memory, and navigation

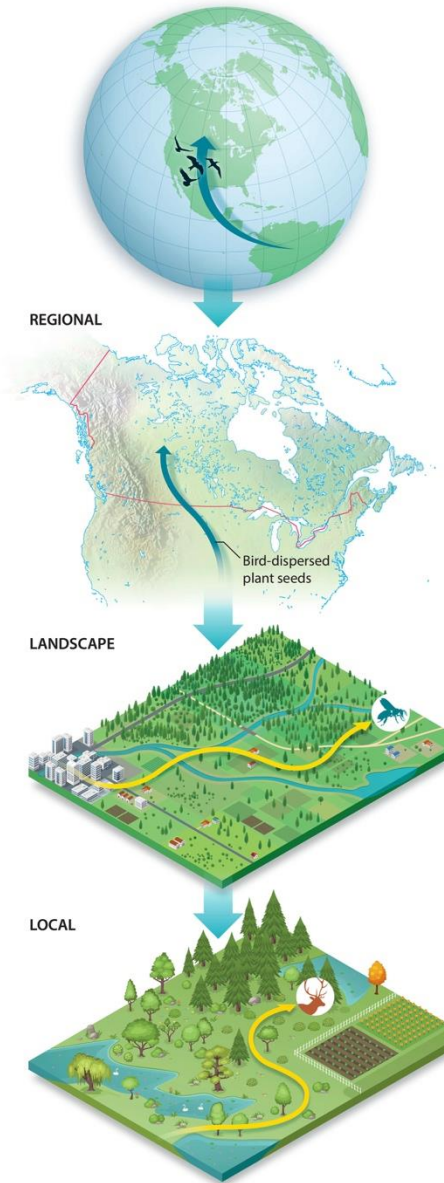
Connectivity

- Can be estimated as structural or functional connectivity
- Conservation efforts to increase connectivity can scale across space and species



Fletcher Jr & Fortin, 2025

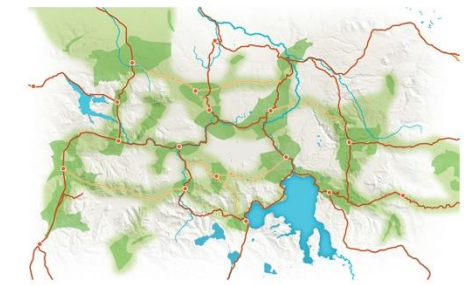
Scale
GLOBAL



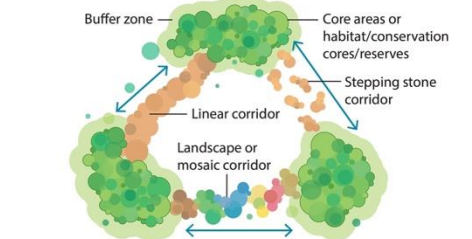
Examples of connectivity application
MIGRATORY PATHWAYS



PROTECTED AREA NETWORKS



HABITAT CORRIDORS



WILDLIFE CROSSINGS



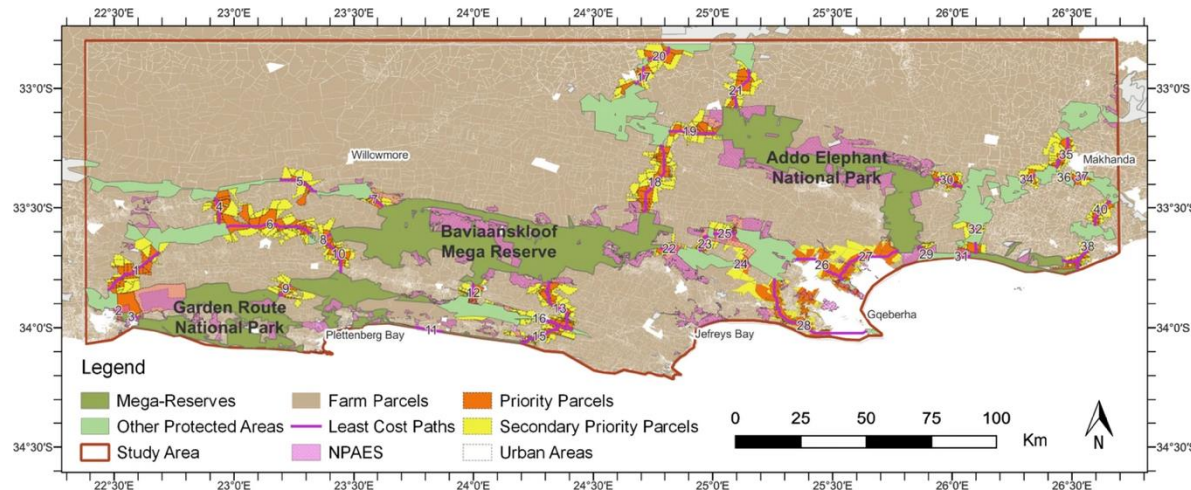
Brodie et al. 2025

Connectivity in action

- Can be modeled to identify priority areas for corridor protection
- Structures can be built to preserve connectivity in urban landscapes



Photo credit: Parks Canada, Allie Banting



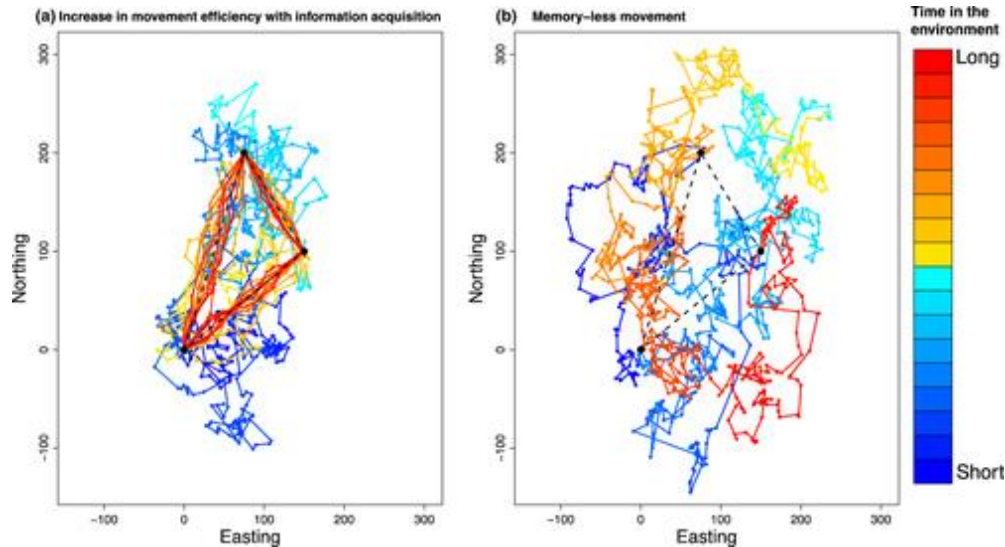
Lichtenberg et al., 2025



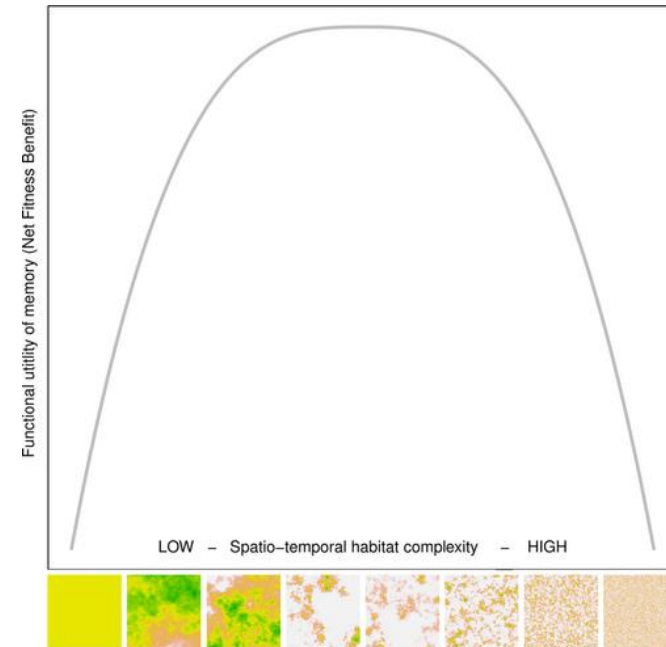
Photo credit: Michael Dantas/WCS

Memory

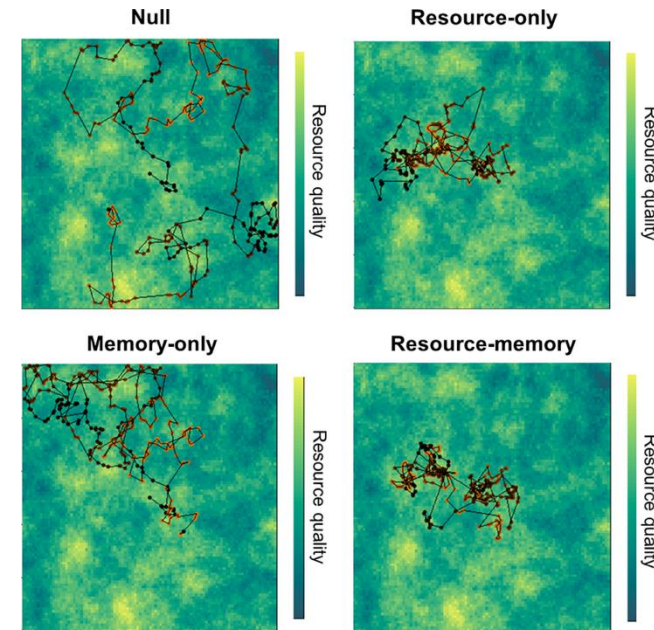
- Considered to be important component of animal ecology, but often not addressed
- Often associated w/ routine behaviors and location use
- Particularly beneficial when the environment is predictable



Fagan et al., 2013



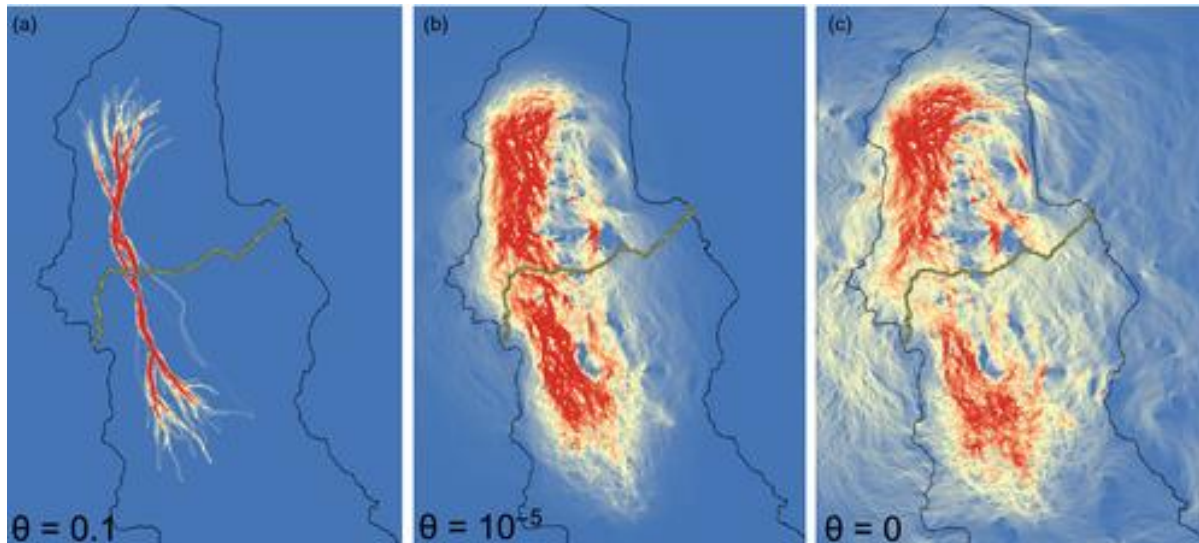
Fagan et al. 2013



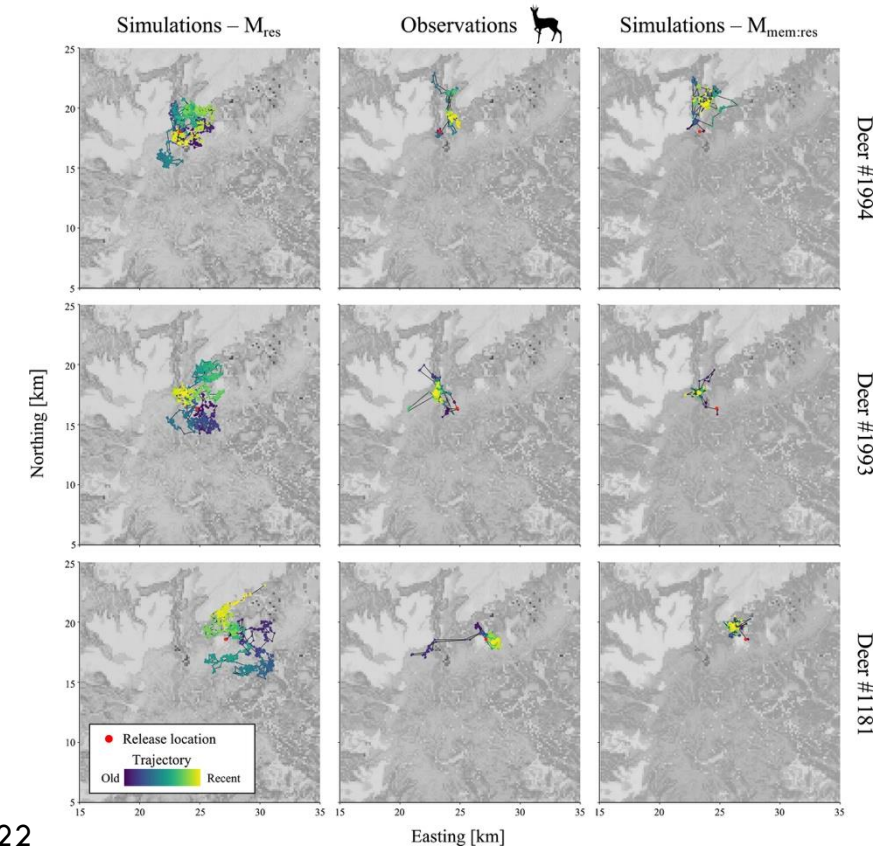
Thompson et al. 2022

Memory in action

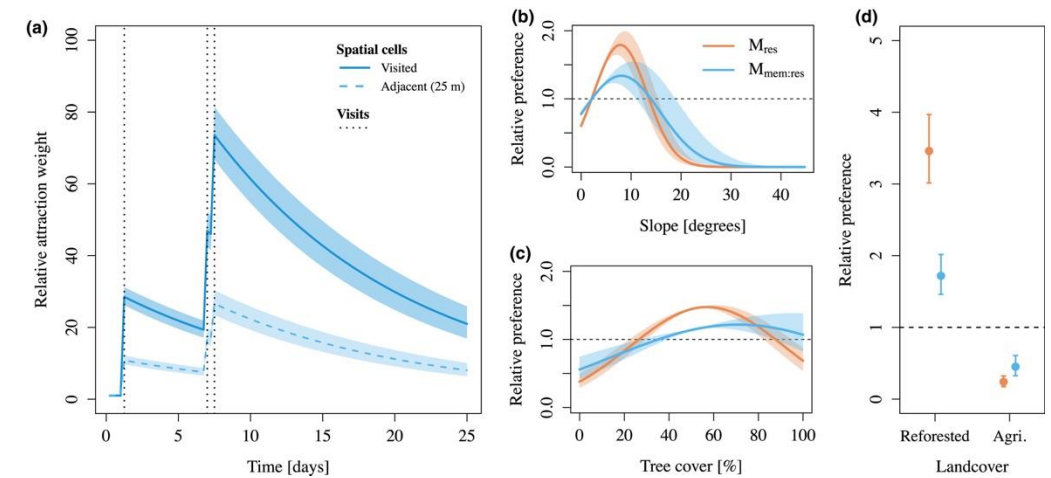
- Can be included in movement and step selection models
- Can be added to models in variety of ways



Panzacchi et al., 2016

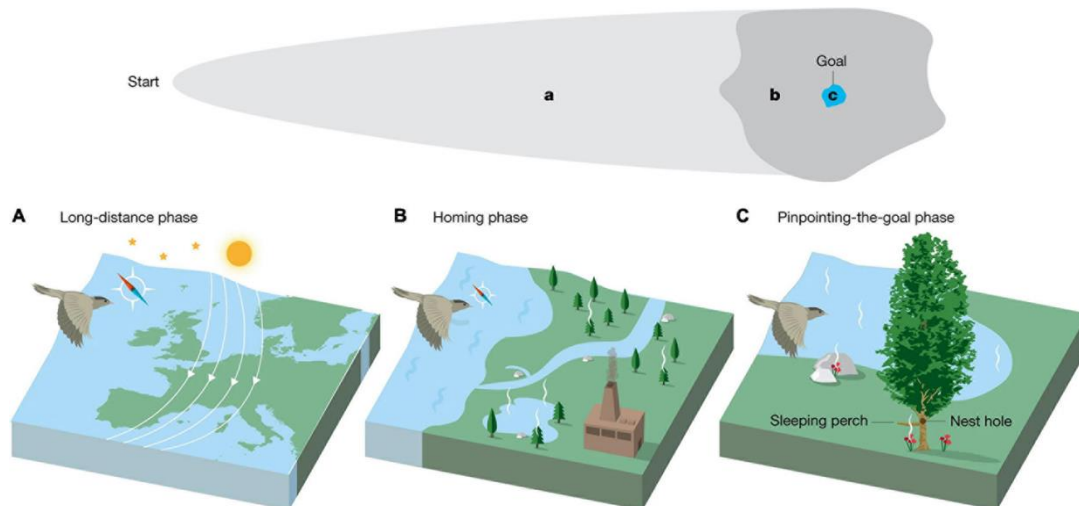


Ranc et al., 2022

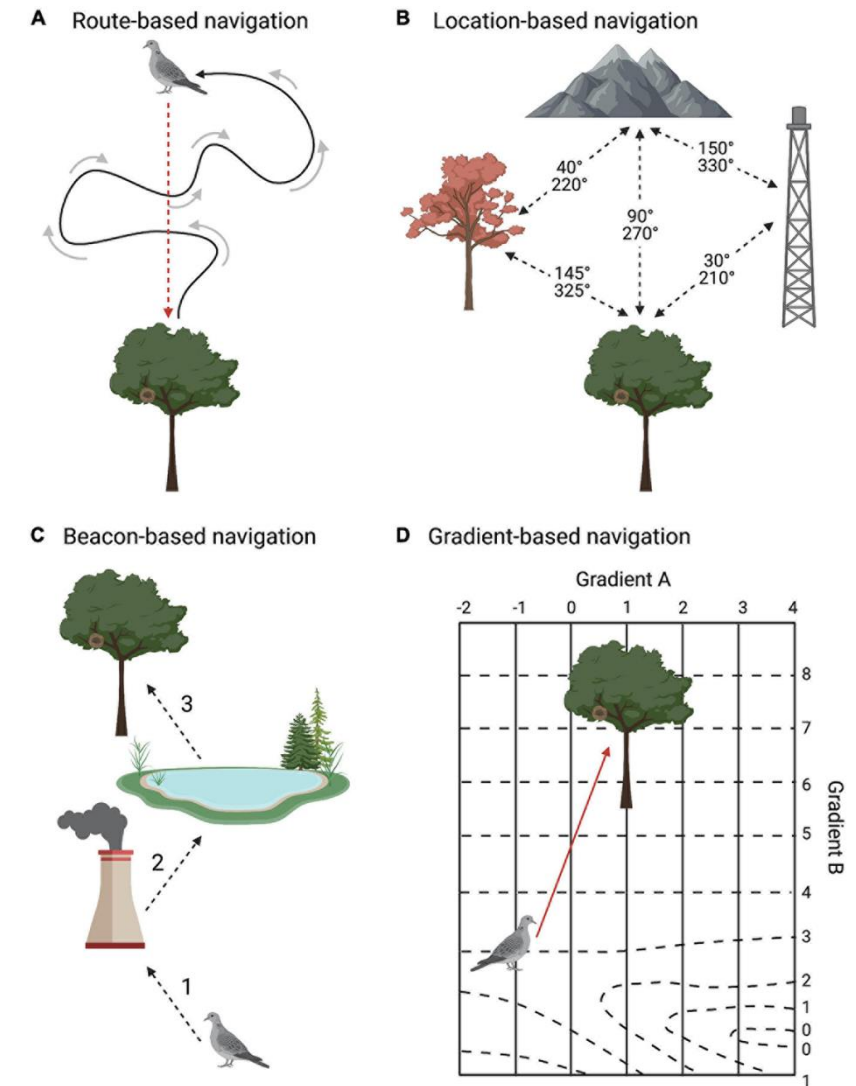


Navigation

- Animals often use a variety of cues during long-distance migrations
 - Includes olfaction, sun position, stars, geomagnetic field, wind/current direction, etc
- These cues may also change as they approach their destination



Mouritsen, 2018



Kashetsky et al. 2021

Time for live-coding

Data cleaning, exploration, and processing

